Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/05/2006
PI Name:	Thorne, Robert E. Ph.D.		
Project Title:	Macromolecular Crystal Growth: Techniques Development, Fundamental Studies, and Applications to Complexes Affecting Cell Growth and Division		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Structural biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ret6@cornell.edu	**Fax:**	FY 607-255-6428
PI Organization Type:	UNIVERSITY	**Phone:**	607-255-6487
Organization Name:	Cornell University		
PI Address 1:	Physics Department, 528 Clark Hall		
PI Address 2:	Cornell University		
PI Web Page:	www.ccmr.cornell.edu/~robt/thorne-group.html; www.ccmr.cornell.edu/~robt/micromounts.html		
City:	Ithaca	**State:**	NY
Zip Code:	14853-2501	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	08/30/2001	**End Date:**	11/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Gorti, Sridhar	**Contact Phone:**	256 544-0158
Contact Email:	sridhar.gorti@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ellenberger, T. (Harvard Medical School) Duffy, T. (Geosciences Princeton University)		
Grant/Contract No.:	NAG8-1831		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Growth of high quality macromolecular crystals is a major factor limiting the rate at which new macromolecular structures are determined. We are pursuing an interdisciplinary experimental program to advance fundamental understanding of macromolecular crystals and to solve specific structures important in cell growth and division. Our program involves two major components: (1) Development and application of novel or improved experimental probes of macromolecular solutions and crystals, including X-ray imaging techniques, SAXS, and Brillouin scattering; and (2) Application of these and other techniques to help obtain high-quality crystals of DNA helicase-primase complexes and CDC42 and its regulators. The objectives of this work are the development of novel or improved experimental probes of macromolecular solutions and crystals, and specific attacks on crystal-related problems of molecular complexes involved in cell growth. New/improved techniques for characterizing macromolecular solutions and crystals should yield		

Task Description:	new insights into intermolecular interactions, crystal disorder, and molecular structure. Applying these and our current techniques (so far restricted to simple model proteins) to macromolecular complexes of significant biological interest will help to identify and refine a subset of these techniques most useful to the structural biology community. The graduate students involved in this program will receive an interdisciplinary training well suited to addressing future challenges in this expanding area. This program will assist NASA's microgravity effort by developing new methodologies for evaluating the success of flight experiments, by testing specific hypotheses about defect formation relevant to the microgravity environment, and by helping to disseminate the results from its microgravity crystal growth program.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work should significantly impact the general practice of macromolecular crystallography, and thus the fundamental understanding of biological processes and the development of new medicines.
Task Progress:	During the last year we continued our studies of contact line pinning of drops. We showed that the irregular pinning and irregular drop shapes that occurs in standard crystallization media leads to large irreproducibilities in protein crystallization, especially in nucleation. In the last year we have explored a variety of microfabrication-based methods to obtain stronger contact line pinning. We have shown that we can dramatically improve the shape reproducibility of drops and the maximum drop volume that remains stable when the drops are flipped. Based on these results, we have designed a new type of crystallization plate. This plate is compatible with existing liquid handling robots, and should yield improved crystallization results and allow easier crystal retrieval and in-situ X-ray examination.
Bibliography Type:	Description: (Last Updated: 03/14/2017)
Articles in Peer-reviewed Journals	Berejnov V, Thorne RE. "Enhancing drop stability in protein crystallization by chemical patterning. " Acta Crystallogr D Biol Crystallogr. 2005 Dec;61(Pt 12):1563-7. PMID: 16301789 , Dec-2005
Articles in Peer-reviewed Journals	Kalinin Y, Thorne R. "Crystal growth in X-ray-transparent plastic tubing: an alternative for high-throughput applications." Acta Crystallogr D Biol Crystallogr. 2005 Nov;61(Pt 11):1528-32. PMID: 16239731 , Nov-2005
Articles in Peer-reviewed Journals	Kalinin,Y, Kmetko J, Bartnik A, Stewart A, Gillilan R, Lobkovsky E, Thorne RE. "A New Sample Mounting Technique for Room-Temperature Macromolecular Crystallography. " J Appl Cryst. 2005 Apr;38(Pt 2):333-9. , Apr-2005
Articles in Peer-reviewed Journals	Malkin AJ, Thorne RE. "Growth and disorder of macromolecular crystals: Insights from atomic force microscopy and X-ray diffraction studies. " Methods. 2004 Nov;34(3):273-99. PMID: 15325647 , Nov-2004
Patents	Cornell Docket D-D-3569. Pending Jul-2005 Thorne, R. E. and Berejnov, V. "Screening and Crystallization Plates for Manual and High-Throughput Protein Crystal Growth."
Patents	Cornell Docket D-3545. Pending Jan-2006 Thorne, R. E. "Microfabricated Tools for Manipulation of Small Samples."